

Finding memory details: GNU/Linux has inbuilt ‘free’ commands, which display the total amount of free and used physical and swap memory in the system. In addition, it also shows the buffers and caches used by the kernel. Given below is the sample output of memory usage in megabytes:

```
$ free -m
      total    used    free   shared  buff/cache   available
Mem:   7923    935    3229     231     3758     6463
Swap:  2047      0    2047
```

In the above command, the `-m` option shows details in megabytes. One can also get the same information in different units, like bytes, kilobytes and gigabytes.

Find the frequency of the RAM: Systems administrators often need to upgrade hardware. To upgrade the RAM, we must know its frequency. Using the commands below, we can find the frequency of the RAM:

```
$ sudo lshw -short -C memory
H/W path    Device      Class        Description
=====
/0/0                               memory       1MiB BIOS
/0/6                               memory       8GiB System
Memory
/0/6/0                               memory       4GiB SODIMM
DDR3 1333 MHz (0.8 ns)
/0/6/1                               memory       4GiB SODIMM
DDR3 1333 MHz (0.8 ns)
/0/c/e                               memory       32KiB L1 cache
/0/c/f                               memory       256KiB L2
cache
/0/c/10                               memory       3MiB L3 cache
/0/d                               memory       32KiB L1 cache
```

 **Note:** This command is executed with root privileges.

Network related commands

Resolving the domain name to the IP address: Using the `nslookup` command, we can find the IP address associated with a particular domain name. For example, the commands below find the IP address of `google.com`:

```
$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name: google.com
Address: 172.217.160.174
Name: google.com
Address: 2404:6800:4009:80a::200e
```

Tracing the route taken by the network packet: Using the `traceroute` command, we can find the number of hops between the source and the destination machines. In addition, we can also trace the path travelled by the packet using this utility. For instance, the output below shows the path travelled by the packet:

```
$ traceroute google.com
traceroute to google.com (216.58.203.206), 30 hops max, 60
byte packets
 1  _gateway (192.168.1.1)  7.073 ms  7.044 ms  7.024 ms
 2  * * *
 3  10.124.117.210 (10.124.117.210)  41.581 ms  41.585 ms
 41.570 ms
 4  10.117.225.82 (10.117.225.82)  149.475 ms  149.525 ms
 149.510 ms
 5  10.117.137.146 (10.117.137.146)  84.640 ms  84.663 ms
 84.649 ms
 6  14.141.63.225.static-Mumbai.vsnl.net.in (14.141.63.225)
 53.287 ms  37.258 ms  39.688 ms
```

 **Note:** Some part of the output is not shown here to save space.

Netstat: The Netstat utility shows network statistics. It prints network connections, routing tables, interface statistics and other useful information. To display routing table information, execute the following commands:

```
$ netstat -r
Kernel IP routing table
Destination Gateway Genmask Flags MSS Window irtt Iface
default _gateway 0.0.0.0 UG 0 0 0 wlp2s0
link-local 0.0.0.0 255.255.0.0 U 0 0 0 wlp2s0
172.16.208.0 0.0.0.0 255.255.255.0 U 0 0 0 vmnet1
192.168.1.0 0.0.0.0 255.255.255.0 U 0 0 0 wlp2s0
192.168.195.0 0.0.0.0 255.255.255.0 U 0 0 0 vmnet8
```

Miscellaneous commands

Generating a random password: Systems administrators often do user management. As a part of this, one has to generate one-time random passwords. We can use the `openssl` utility to generate random passwords, as follows:

```
$ openssl rand -base64 12
```

Making a file immutable: Some files are so important that we cannot afford to lose them even by accident. GNU/Linux provides the `chattr` command, with which we can make a file immutable. Once a file is marked as immutable, it cannot be deleted even by the root user. Let us look at this with an example:

```
$ touch immutable.txt
$ sudo chattr +i immutable.txt
```

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